

NAVIN KUMAR M

Greater Chennai Area, India

[PhoneNo](#) | [Email](#) | [LinkedIn](#) | [Github](#) | [Website](#)

ABOUT ME

Machine Learning Engineer with 2+ years of experience building scalable AI systems for credit risk and enterprise decision intelligence. A pragmatic problem solver with a strong passion for applying AI to deliver impactful solutions in fintech and enterprise domains.

EDUCATION

B.Tech in Computer Science & Engineering, Specialization in Artificial Intelligence & Machine Learning Sep 2020 - May 2024
Vellore Institute of Technology, India
GPA: 8.92/10

Achievements: Best Capstone Award 2024, VIT University, for developing an on-premise AI learning platform.
Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Structures & Algorithms, Operating System, Networks & Security, DBMS, Linear Algebra, Statistics, Software Engineering, Product Development & Entrepreneurship.

LEADERSHIP ROLES

Co-Founder, Chief Technology Officer, Stealth AI Startup 2025 - Present

- Led a team of 5 in developing tech infrastructure, an AI-driven search, recommendation analytics, and an intelligent document processing engine.
- Drove expansion to 50+ enterprise clients by integrating customer feedback, refining AI models, and ensuring product-market fit.

Project Lead, Machine Learning Initiatives, Branch International 2024 - Present

- Led the development of several credit risk models for underwriting that unlocked \$100M+ in responsible financial access across India and Africa, enabling data-driven lending for 50M+ underserved users.
- Strengthened AI-driven lending systems through reliable training and inference workflows, supported by cross-functional collaboration and multiple initiatives that elevated model quality and deployment standards.

WORK EXPERIENCE

Machine Learning Engineer, Branch International, Remote Jun 2024 – Present

- Drove the development of a neural network-based credit risk model from unstructured SMS data, ensuring transparent and interpretable underwriting decisions aligned with regulatory standards.
- Accelerated feature fetch processing by 8 times through a distributed head-worker system using Ray, fine-tuning service worker and threadpool configurations, optimizing SQL queries, and intelligent batching of I/O- & CPU- intensive requests. [\(blog\)](#)
- Developed a credit model using bureau data for the premium user segment, achieving a 3% delinquency rate, at 29% APR reduction.
- Built a fully automated, scalable training infrastructure on AWS, reducing model training time and costs by fivefold.
- Integrated a RabbitMQ-based priority queue system to orchestrate reliable communication between backend and AI services.
- Led the effort of integrating bureau and account aggregator data sources into our data & feature engineering pipeline
- Decreased feature service latency by threefold by rewriting critical CPU-intensive components in Rust
- Refined training sample queries and incorporated shorter window targets, enabling the use of recent data along with improving data confidence & generalizability across all user segments.

INTERNSHIPS

Machine Learning Engineer, Branch International, Remote Jan 2024 – May 2024

- Trained robust BERT-based cashflow classification & extraction SMS model with AUC of 0.99, increasing data coverage threefold, and maintained year-long operational stability.
- Engineered bureau-based features and trained a credit model for first-time borrowers, resulting in a 6% AUC improvement and enabling \$90 million in financial access for underserved users.
- Developed a parallel processing algorithm to identify issues in retrospective bureau data and transform it into Credit Reports at 100K+ files/min.

Engineering Intern, Vellore Institute of Technology, India Nov 2023 – Apr 2024

- Built a parallel and distributed computing workspace using Ray, NFS to accelerate ML training and inference.
- Configured Linux & ML Frameworks on IBM PowerPC ppc64le architecture.
- Architected and deployed an AI Learning Platform on distributed clusters to support hands-on learning for junior students.

Machine Learning Engineer, Gutsy Innovation, Remote May 2023 – July 2024

- Built a real-time face detection & recognition system using YOLOv7 and ArcFace.
- Integrated Kalman filter-based face tracking and ESPCN super resolution to enhance image clarity.
- Deployed this AI surveillance service in edge devices using DeepStream SDK and integrated it with Azure IoT Hub for monitoring.

ACADEMIC PROJECTS

Instruction-Aligned AI Tutor, Vellore Institute of Technology, India [\(github\)](#) Nov 2023 – Apr 2024
Role: Individual Contributor **Team Size:** 1

- Fine-tuned LLaMA-7B using RLHF on synthetic Q&A corpora with adversarial prompts, improving pedagogical alignment and mitigating prompt injection vulnerabilities.
- Engineered a high-throughput LLM-RAG system using vLLM and Ray, integrated with Qdrant and LangChain for contextual, real-time doubt clarification.
- Built a scalable Django backend with PostgreSQL and Cassandra to retain user interactions for continuous retraining and future model iterations.

Multimodal Malware Forensics , Vellore Institute of Technology, India
[\(github\)](#)

Dec 2022 – Apr 2023

Role: Project Lead

Team Size: 4

- Trained a CoAtNet Transformer on 2M malware binaries converted into dual-channel images using byteplot and bigram-DCT transforms to learn morphological malware signatures.
- Trained LightGBM models on static PE structural attributes extracted from binaries, and built an LSTM classifier over dynamic system-call traces obtained from sandbox execution logs.
- Ensembled all classifiers using a meta-model for production inference, and developed a cross-platform Rust-based client that uploads files to an AI-driven backend with real-time model scoring visualized via Azure Power BI.

Spatiotemporal Anomaly Perception, Vellore Institute of Technology, India
[\(github\)](#)

Dec 2022 – Apr 2023

Role: Project Lead

Team Size: 4

- Trained a SwinTransformer as a variational autoencoder for frame reconstruction, and fused its encoder with LSTM decoders to capture temporal anomalies in UCF-Crime surveillance streams.
- Optimized anomaly classification by adding an SVM-based prefilter, calibrated to capture minute anomalies, which is then verified by an LSTM network to reduce computation and false positives.
- Deployed an edge-ready WebSocket pipeline for multi-camera ingestion and low-latency anomaly alerts.

Pix2Struct-Based Vision-Language Graph Parsing, Vellore Institute of Technology, India
[\(github\)](#)

Apr 2023

Role: Individual Contributor

Team Size: 1

- Performed visual reasoning on charts and plots to make them accessible for visually impaired users.
- Used EfficientNetV2 for chart classification and applied Matcha (Pix2Struct) for bar, line, and dot plot interpretation, along with Faster R-CNN for scatter plot derendering.

ACHIEVEMENTS/AWARDS/CO-CURRICULAR/EXTRACURRICULAR

• Winner, Branch International - Hall of Fame Award for Best Performing Engineer	(award)	Oct 2025
• Runner, Branch Hackathon – For Building a GenAI Credit Risk Model	(award)	Jan 2025
• Winner, Best Final Year Project, VIT University, for developing an on-premise AI learning platform	(award)	Apr 2024
• Winner, Branch Hackathon – For developing an AI Agent to assist engineers with documentation	(award)	Jan 2024
• 98.92%ile, in Jee Mains, conducted all over India		Jan 2020
• Top 1% in 10th Grade Board Exams		Jul 2018
• Represented Inter-Club Cricket Team as a Left-Arm Fast Bowler & Captain		2014 - 2017
• Piano Enthusiast and Performer		2012 - 2017

TECHNICAL SKILLS

Technical Expertise:	Natural Language Processing (4 yrs), Cloud Infrastructure (4 yrs), MLOps (3 yrs), Credit & Fraud Modeling (2 yrs), Generative Language Models (1 yrs), Forecasting Modeling (2 yrs), Computer Vision (1 yr), Information Security (1 yr)
Cloud & Infrastructure:	AWS (4 yrs), Git (4 yrs), Ray (3 yrs), Linux (3yrs), Docker & Kubernetes (2 yr), Airflow (1 yrs)
Programming Languages:	Python (6 yrs), C++ (4 yrs), SQL (2 yrs), Rust (1 yr), Go (1 yr), Cuda (1 yr)
Frameworks & Tools	PyTorch (4 yrs), Scikit (3 yrs), Triton (1 yr), Power BI (1 yr), Matlab (1 yr)

CERTIFICATIONS/WORKSHOPS/ADDITIONAL COURSES

• Machine Learning in Production, Coursera – offered by DeepLearning.AI	(certification)	Jul 2023
• AWS Certified Developer – Associate, Udemy – preparation course for AWS	(certification)	May 2023

COMMUNITY INVOLVEMENT

Public Relations Officer – Rotaract Club, Vellore Institute Of Technology

Jul 2022 – Dec 2023

- Spearheaded communication, outreach, and marketing efforts for the club.
- Organized fundraising events, coordinated donation drives for underprivileged communities,
- Played a key role in the execution of events such as blood donation camps, cleanliness drives, and educational workshops.

LANGUAGES KNOWN

Tamil (native), English (C1), German (A1).